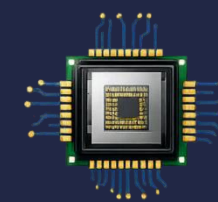
CS-301 Microprocessor-Based System Design



Lecture 28: Interrupts

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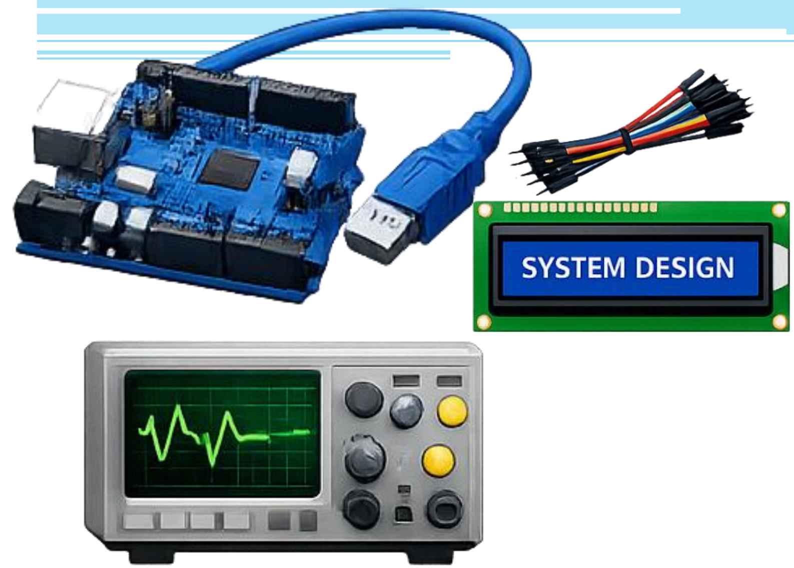

Interrupt-Based Timer0

Design application:

- Uses ***Timer0 in normal mode***
- Generates periodic interrupts using overflow
- Toggles an LED every fixed interval
- Demonstrates:
    - Timer initialization
    - ISR creation
    - Global interrupt enabling
    - Overflow counting

Timer0 in the ATmega162 is an 8-bit hardware timer/counter built into the microcontroller
A periodic interrupt is an interrupt that occurs repeatedly at fixed time intervals..
The CPU temporarily pauses the current program.
It executes the Interrupt Service Routine (ISR).
After the ISR finishes, the CPU resumes the interrupted program.

Interrupt-Based Timer/Counter0

Design application:

MCU clock = 8 MHz
LED connected to PB0
Timer0 prescaler = 1024
We will blink LED approximately every 1 second.
So, 8 MHz /1024= 7812.5 Hz
This means 7812.5 clock ticks in 1 sec
⇒ 1 clock tick will take 1/7812.5 sec
⇒ 256 clock ticks will take 32.768msec
⇒ 1 overflow will take 32.768 msec
⇒ 30.5 overflows will take around 1000 msec
⇒ So the number of times overflow interrupt should occur must be around 31.

Some Vector Names

Refer to Page 59 of ATMEGA162 datasheet

Interrupt Source	**Vector Name**
External INT0	INT0_vect
Timer0 Overflow	TIMER0_OVF_vect
Timer1 Overflow	TIMER1_OVF_vect
USART RX	USART0_RX_vect